

Lecture 5 – Distributed Hash Tables & Chord (25 MCQs)

1. A Distributed Hash Table (DHT) is best described as:

- A) A centralized database
- B) A hash table implemented on a single server
- C) A distributed abstraction of a hash table
- D) A file system protocol

Answer: C

2. In a DHT, a key is typically generated using:

- A) File name only
- B) Node IP address
- C) A hash function applied to data
- D) Network routing tables

Answer: C

3. The main problem a DHT is designed to solve is:

- A) Data encryption
- B) Load balancing
- C) Data lookup in large distributed systems
- D) Fault detection

Answer: C

4. Sharding in distributed systems refers to:

- A) Replicating data everywhere
- B) Splitting data into independent pieces
- C) Encrypting data before storage
- D) Centralizing storage

Answer: B

5. A key property of DHT systems is that they are:

- A) Centralized
- B) Static
- C) Decentralized
- D) Hierarchical

Answer: C

6. Why is a central lookup server avoided in DHTs?

- A) It increases performance
- B) It improves scalability
- C) It introduces a single point of failure
- D) It simplifies routing

Answer: C

7. If each peer stored only its own data, a major issue would be:

- A) Faster lookup
- B) Better load balancing
- C) Data loss when a peer leaves
- D) Reduced replication overhead

Answer: C

8. Chord improves over naïve peer storage by:

- A) Centralizing control
- B) Using consistent hashing
- C) Broadcasting requests
- D) Eliminating hashing

Answer: B

9. Consistent hashing is mainly used to:

- A) Encrypt data
- B) Map keys to nodes evenly
- C) Increase latency
- D) Prevent replication

Answer: B

10. In Chord, both nodes and data items are:

- A) Assigned sequential IDs
- B) Identified using the same hash function
- C) Stored at random locations
- D) Managed by a central authority

Answer: B

11. The identifier space in Chord is:

- A) Linear
- B) Tree-based
- C) Ring-shaped
- D) Star-shaped

Answer: C

12. In Chord, a data item with key k is stored at:

- A) The node with the same ID
- B) The predecessor of k
- C) The successor node with $ID \geq k$
- D) The node that uploaded it

Answer: C

13. The primary goal of consistent hashing is to achieve:

- A) Centralized management
- B) Load balancing
- C) Strong consistency
- D) Data encryption

Answer: B

14. A major advantage of Chord is that lookup operations require:

- A) $O(1)$ hops
- B) $O(N)$ hops
- C) $O(\log N)$ hops
- D) $O(N^2)$ hops

Answer: C

15. The main reason for using finger tables in Chord is to:

- A) Store data replicas
- B) Maintain consistency
- C) Speed up lookup operations
- D) Authenticate nodes

Answer: C

16. Each node in Chord maintains a finger table of size:

- A) N entries
- B) $\log N$ entries
- C) m entries, where m is identifier bits
- D) 2^m entries

Answer: C

17. Finger tables contain:

- A) All nodes in the network
- B) Shortcuts to selected nodes in the ring
- C) Copies of all keys
- D) Only successor nodes

Answer: B

18. A basic Chord lookup without finger tables would require:

- A) $O(\log N)$ hops
- B) $O(1)$ hops
- C) $O(N)$ hops
- D) Constant time

Answer: C

19. One reason Chord hashes node addresses is to:

- A) Improve security
- B) Reduce communication cost
- C) Achieve uniform distribution and independent failures
- D) Simplify routing logic

Answer: C

20. Chord is considered dynamic because:

- A) Nodes never fail
- B) Nodes can join and leave at any time
- C) Data never moves
- D) The ring is static

Answer: B

21. Replication in Chord helps to:

- A) Eliminate hashing
- B) Reduce lookup time

- C) Improve fault tolerance
- D) Centralize data

Answer: C

22. Which of the following is NOT a desired property of a DHT?

- A) Scalability
- B) Decentralization
- C) Low lookup latency
- D) Central authority

Answer: D

23. In Chord, load balancing is achieved because:

- A) Each node stores its own data only
- B) Data is assigned based on upload location
- C) Hashing distributes keys uniformly
- D) Nodes choose their data

Answer: C

24. The lookup(key) operation in a DHT returns:

- A) The data itself
- B) The IP address of the responsible node
- C) All nodes storing replicas
- D) The predecessor node

Answer: B

25. Compared to centralized indexing, Chord-based DHTs are:

- A) Less scalable
- B) More vulnerable to failures
- C) More scalable and fault tolerant
- D) Slower for large systems

Answer: C